



COURSE SYLLABUS

CS 3331 – Fundamentals of Artificial Intelligence

INSTRUCTOR INFORMATION

Name: Dr. Pham Huu Du
Email: du.phamhuu@hust.edu.vn

COURSE DESCRIPTION:

This course will introduce the basic ideas and techniques of artificial intelligence: intelligent agents, search strategies, constraint satisfaction, logic and automatic proofing, knowledge representation, uncertain knowledge and reasoning. By doing a capstone project at the end of this course, students will gain practical experience in building an AI system. In addition, students will practice necessary skills for future work such as teamwork skills, research skills, writing reports and presentations.

Prerequisites: CS 3323 or CS 3330.

OVERALL OBJECTIVES:

After this course the student will obtain the following:

- ✧ Understand basic concepts and techniques of AI
- ✧ Understand advanced concepts and techniques in AI: uncertain knowledge and reasoning, machine learning
- ✧ Be able to apply the learned knowledge to build intelligent software through assignment in groups
- ✧ Be able to identify research areas and potential developments of artificial intelligence

COURSE TEXTBOOKS:

- [1] Russell and P. Norvig. *Artificial Intelligence: A Modern Approach* (3rd Edition). Prentice Hall, 2009.
- [2] T. M. Mitchell. *Machine Learning*. McGraw-Hill, 1997.

COURSE MATERIALS:

- Material class on MS. Teams/ Website is available for students to get all course materials, announcements, submit homeworks, ...

Grading Scale

90-100 - A

80- 89 - B

70- 79 - C

60- 69 - D

Below 60 - F

Grading Computation:

Homeworks: 10%

Group project: 30%

Midterm test: 20%

Final Exam: 40%

ATTENDANCE

Student's attendance will be marked for every lesson. A student who is absent more than **03 times** with or without faculty's permission will get F for the course.

SUBMITTING HOMEWORKS

The homework will be submitted to instructor via MS. Teams

LATE POLICY

Each homework has a deadline as noticed at class. Any homework turned in after its deadline will be not accepted.

COURSE SCHEDULE

Week	Chapter	Topic
1	1	Chapter 1: Introduction 1.1. What is AI? 1.2. The foundation of AI 1.3. The history of AI 1.4. Research and application areas of AI 1.5. Future of AI
2	2	Chapter 2: Intelligent agents
3	3	Chapter 3: Problem solving 3.1. Solving problem by searching
4	3	3.1. Solving problem by searching (con't) and A*
5	3	3.2. Adversarial search
6	3	3.4. Constraint satisfaction problems

7	3	Homework #1
8	4	Midterm Exam
9	4	Chapter 4: Knowledge and Inference 4.1. Knowledge representation
10	4	4.2. Propositional logic 4.3. Inference in propositional logic
11	4	4.4. First-order logic
12	4	4.5. Inference in first-order logic Homework #2
13	5	Chapter 5: Machine learning and Advanced AI
14		Project defending
15		Final Exam